Appendix A AJAE Appendix for Measurement Error Mechanisms Matter: Full Derivations From Section 2

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The material contained herein is supplementary to the article named in the title and published in the *American Journal of Agricultural Economics*.

In this portion of the appendix, we repeat the equations reported in the main text – keeping the same equation numbers for ease of comparison – and provide the algebraic derivations of the results featured in the text. The two main equations of interest are:

$$Y - X = \beta^o X + \varepsilon \tag{1}$$

$$Y - X^* = \beta^* X^* + \epsilon \tag{2}$$

And measurement error follows:

$$X = X^* + v \tag{3}$$

We assume ϵ is orthogonal to X^*, and ε is orthogonal to both X^* and v.

Scenario 1: Misreporting

In the main text, we explain the misreporting scenario as it might arise from a global regression-to-mean or a focal-point-bunching process.[24] We assume:

$$v = \alpha X^* + \psi \tag{4}$$

Where ψ is assumed to be orthogonal to X^* and to ϵ. It follows that:

$$X = X^* + v = X^* + \alpha X^* + \psi = (1 + \alpha)X^* + \psi \tag{5}$$

Since the data generating process follows equation 2, we can rearrange equation 1 as:

$$Y - X = Y - X^* - v = (\beta^* X^* + \epsilon) - (\alpha X^* + \psi) = \beta^* (X - v) - \alpha X^* + \epsilon - \psi$$

$$= \beta^* X - \beta^* v - [\alpha X^* + \psi] + \epsilon$$

$$= \beta^* X - \beta^* v - v + \epsilon$$

$$= \beta^* X - (\beta^* + 1)v + \epsilon$$

Therefore, estimating equation 1 will generate the biased coefficient estimate:

$$\widehat{\beta^o} = \frac{Cov([Y - X], X)}{Var(X)} = \frac{Cov([\beta^* X - (\beta^* + 1)v + \epsilon], X)}{Var(X)} = \frac{Cov([\beta^* X - (\beta^* + 1)v], X)}{Var(X)}$$

[24] Focal point bunching reflects local regression-to-the-mean around focal points but is necessarily non-linear and non-monotone so not directly reflected in equation 4.

$$\Rightarrow \widehat{\beta^o} = \beta^* - (\beta^* + 1)\frac{Cov\,(v, X)}{Var\,(X)} \tag{6}$$

And this can be further specified in terms of the data generating process parameters:

$$\Rightarrow \widehat{\beta^o} = \beta^* - (\beta^* + 1)\frac{Cov\big(\alpha X^* + \psi, (1+\alpha)X^* + \psi\big)}{Var\big((1+\alpha)X^* + \psi\big)}$$

$$= \beta^* - (\beta^* + 1)\left[\frac{\alpha(1+\alpha)\sigma^2_{X^*}}{(1+\alpha)^2\sigma^2_{X^*} + \sigma^2_\psi} + \frac{\sigma^2_\psi}{(1+\alpha)^2\sigma^2_{X^*} + \sigma^2_\psi}\right]$$

Alternatively, $\widehat{\beta^o}$ can be derived as a function of $Cov\,(v, X^*)$ rather than of $Cov\,(v, X)$:

$$\widehat{\beta^o} = \frac{Cov\big([Y - X], X\big)}{Var\big(X\big)} = \frac{Cov\big([(\beta^* + 1)X^* + \epsilon - (X^* + v)], X^* + v\big)}{Var\big(X^* + v\big)}$$

$$= \frac{Cov\big([\beta^* X^* + \epsilon - v], X^* + v\big)}{Var\big(X^* + v\big)} = \frac{Cov\big([\beta^* X^* - v], X^* + v\big)}{Var\big(X^* + v\big)}$$

$$\Rightarrow \widehat{\beta^o} = \frac{\beta^*\sigma^2_{x^*} + (\beta^* - 1)\,Cov\,(v, X^*) - \sigma^2_v}{\sigma^2_{x^*} + \sigma^2_v + 2Cov\,(v, X^*)} = \beta^*\frac{Cov\,(X^*, X^* + v)}{Var\,(X^* + v)} - \frac{Cov\,(v, X^* + v)}{Var\,(X^* + v)}$$

An alternative mechanism behind misreporting might arise from optimal prediction error (Hyslop and Imbens, 2001). For instance, if seeding density or fertilizer application rate recommendations are common knowledge, then farmers may report the plot size that corresponds to these recommendations, given the exact quantity of seed or fertilizer they know they recently purchased and fully applied to the plot. As we show, this process generates a similar form of non-classical measurement error, and implies a similar form of bias in the intensification parameter of interest. That is, distinct data generating processes driven by pure misreporting result in similar outcomes.

To illustrate this, let us now assume a different data misreporting process, one in which NCME reflects a relationship between measurement error v and input quantity Y:

$$v = \delta\, Y + \omega$$

where ω is assumed uncorrelated with X^* and with ϵ from equation 2. Combining equations 2 and 7 yields $v = \delta\,(\beta^* + 1)\,X^* + \delta\epsilon + \omega$. By defining $\kappa \equiv \delta\,(\beta^* + 1)$ we can define the functional form of measurement error as

$$v = \kappa X^* + \delta\epsilon + \omega$$

If $\kappa < 0$, measurement error again drives regression to the mean, just as occurs if $\alpha < 0$ under scenario 1. Although the precise mechanisms behind the NCME differ, a common reduced form relationship exists. Combining expressions results in

$$X = (1 + \kappa)X^* + \delta\epsilon + \omega$$

These equations are remarkably similar to those from the misreporting case in the main text, despite the difference in the data generating process underlying the measurement error. Measurement error remains a reduced form function of true plot size, but now with a second term that reflects the component of input levels not derived from plot size, ϵ , reflecting that farmer beliefs about plot size are formed based on observations of Y. This leads to an extra term in the variance of self-reported plot size.

Under this form of NCME, OLS estimation of equation 2 (i.e., using GPS-measured plot size), will again result in an unbiased estimate of β^*. Since again the data generating process follows equation 2, we can derive:

$$
\begin{aligned}
&Y - X = Y - X^* - v \\
&= (\beta^* X^* + \epsilon) - (\kappa X^* + \delta\epsilon + \omega) \\
&= (\beta^* - \kappa)X^* + (1-\delta)\epsilon - \omega \\
&= \beta^* (X - v) - \kappa X^* + \epsilon - \delta\epsilon - \omega \\
&= \beta^* X - \beta^* v - [\kappa X^* + \delta\epsilon + \omega] + \epsilon \\
&= \beta^* X - \beta^* v - v + \epsilon \\
&= \beta^* X - (\beta^* + 1)v + \epsilon
\end{aligned}
$$

Therefore estimating equation 1 using self-reported plot size will yield the biased coefficient estimate

$$
\widehat{\beta^o} = \frac{Cov\big([Y - X], X\big)}{Var(X)} = \frac{Cov\big([\beta^* X - (\beta^* + 1)v + \epsilon], X\big)}{Var(X)} = \frac{Cov\big([\beta^* X - (\beta^* + 1)v], X\big)}{Var(X)}
$$

$$
\Rightarrow \widehat{\beta^o} = \beta^* - (\beta^* + 1)\frac{Cov\,(v, X)}{Var\,(X)}
$$

And this can be further specified in terms of the data generating process parameters:

$$
\Rightarrow \widehat{\beta^o} = \beta^* - (\beta^* + 1)\frac{Cov\big(\kappa X^* + \delta\epsilon + \omega, (1+\kappa)X^* + \delta\epsilon + \omega\big)}{Var\big((1+\kappa)X^* + \psi\big)}
$$

$$
= \beta^* - (\beta^* + 1)\left[\frac{\kappa(1+\kappa)\sigma^2_{X^*}}{(1+\kappa)^2\sigma^2_{X^*} + \sigma^2_\omega} + \frac{\delta^2\sigma^2_\epsilon}{(1+\kappa)^2\sigma^2_{X^*} + \sigma^2_\omega} + \frac{\sigma^2_\omega}{(1+\kappa)^2\sigma^2_{X^*} + \sigma^2_\omega}\right]
$$

The general form of bias here is identical to that of equation 6, though the structural parameters within the brackets differ under this new form of measurement error. Under classical measurement error, where $\delta = \kappa = 0$ the first two terms of the bracketed expression again disappear. The last term again reflects division bias arising through the dependent variable, and remains even under classical measurement error.

Scenario 2: Misperceptions

Now we assume that the data generating process follows equation 1. Combining equations 1 and 3 leads to:

$$
Y - X = \beta^o X + \varepsilon = \beta^o(X^* + v) + \varepsilon = \beta^o X^* + \beta^o v + \varepsilon \tag{7}
$$

From equation 7 we can define $Y = \beta^o X^* + \beta^o v + \varepsilon + X$. The outcome variable in equation 2 can therefore be reformulated as:

$$Y - X^* = [\beta^o X^* + \beta^o v + \varepsilon + X] - X^*$$

$$= [\beta^o X^* + \beta^o v + \varepsilon + (X^* + v)] - X^*$$

$$= \beta^o X^* + (\beta^o + 1)v + \varepsilon$$

Estimating equation 2 therefore yields:

$$\widehat{\beta^*} = \frac{Cov\big([Y - X^*], X^*\big)}{Var(X^*)} = \frac{Cov\big([\beta^o X^* + (\beta^o + 1)v + \varepsilon], X^*\big)}{Var(X^*)}$$

$$\Rightarrow \widehat{\beta^*} = \beta^o + (\beta^o + 1)\,\frac{Cov\,(v, X^*)}{Var\,(X^*)} \tag{8}$$

Equation 7 can also be recast so as not to use the self-reported plot size data, to yield:

$$Y - X = \beta^o X^* + \beta^o v + \varepsilon$$

$$\Rightarrow Y - (X^* + v) = \beta^o (X^* + v) + \varepsilon$$

$$\Rightarrow Y - X^* = \beta^o X^* + (\beta^o + 1)v + \varepsilon \tag{9}$$

Scenario 3: Misreporting and Misperception

In this scenario $\widetilde{X}$ is a weighted combination of X and X^* whereby $0 \leq \theta \leq 1$:

$$\widetilde{X} = \theta X + (1 - \theta)\,X^* = \theta v + X^* \tag{10}$$

Therefore a new data-generating process is found:

$$Y - \widetilde{X} = \beta^{\triangle} \widetilde{X} + u \tag{11}$$

where we assume u is orthogonal to both X^* and v. Equation 10 implies $X^* = \widetilde{X} - \theta v$, and equation 11 implies that $Y = (\beta^{\triangle} + 1)\widetilde{X} + u$. By combining equations 3 and 10 we find that:

$$\widetilde{X} = \theta v + X^* = \theta v + (X - v) = (\theta - 1)\,v + X = X - (1 - \theta)\,v$$

or alternatively, $X = \widetilde{X} + (1 - \theta)\,v$. Therefore, equation 1 can be rearranged to take the form

$$Y - X = \Big[\big(\beta^{\triangle} + 1\big)\,\widetilde{X} + u\Big] - \Big[\widetilde{X} + (1 - \theta)\,v\Big]$$

$$= \beta^{\triangle} \widetilde{X} - (1 - \theta)\,v + u$$

$$= \beta^{\triangle}\,(X - (1 - \theta)\,v) - (1 - \theta)\,v + u$$

$$= \beta^{\triangle} X - (\beta^{\triangle} + 1)\,(1 - \theta)\,v + u$$

And so estimating equation 1 will lead to:

$$\widehat{\beta^o} = \frac{Cov([Y - X], X)}{Var(X)} = \frac{Cov([\beta^{\triangle} X - \left(\beta^{\triangle} + 1\right)(1 - \theta)\, v + u], X)}{Var(X)}$$

$$\Rightarrow \widehat{\beta^o} = \beta^{\triangle} - \left(\beta^{\triangle} + 1\right)(1 - \theta)\, \frac{Cov\,(v, X)}{Var\,(X)} \tag{12}$$

Estimating equation 2 will therefore lead to:

$$\widehat{\beta^*} = \frac{Cov\big([Y - X^*], X^*\big)}{Var(X^*)} = \frac{Cov\big([\beta^{\triangle} X^* + (\beta^{\triangle} + 1)\theta v + u], X^*\big)}{Var(X^*)}$$

$$= \frac{Cov\big([\beta^{\triangle} X^* + (\beta^{\triangle} + 1)\theta v], X^*\big)}{Var(X^*)}$$

$$\Rightarrow \widehat{\beta^*} = \beta^{\triangle} + (\beta^{\triangle} + 1)\theta \frac{Cov\,(v, X^*)}{Var\,(X^*)} \tag{13}$$

Equation 2 may also be manipulated to show:

$$Y - X^* = [(\beta^{\triangle} + 1)\widetilde{X} + u] - [\widetilde{X} - \theta v]$$

$$= \beta^{\triangle} \widetilde{X} + u + \theta v$$

$$= \beta^{\triangle}\,(\theta v + X^*) + u + \theta v$$

$$= \beta^{\triangle} X^* + (\beta^{\triangle} + 1)\theta v + u \tag{14}$$

Appendix B AJAE Appendix for Measurement Error Mechanisms Matter: Additional Results

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The material contained herein is supplementary to the article named in the title and published in the *American Journal of Agricultural Economics.*

Table A1 Correlates of measurement error in plot size

	(1) Ethiopia	(2) Malawi	(3) Tanzania	(4) Uganda
GPS-measured plot size (log acres)	-0.323***	-0.506***	-0.396***	-0.426***
	(0.010)	(0.011)	(0.0452)	(0.0619)
Head is male (binary)	-0.043	0.389***	-0.0996**	-0.0449
	(0.078)	(0.026)	(0.0619)	(0.0452)
Age of head (years)	0.000	-0.063***	0.00124	0.000360
	(0.002)	(0.001)	(0.00118)	(0.00197)
Head is literate (binary)	-0.019	-1.198***	0.0317	0.0154
	(0.029)	(0.057)	(0.0361)	(0.0330)
Household size (# members)	0.050***	0.250***	0.00532	-0.0179**
	(0.008)	(0.005)	(0.00470)	(0.00896)
Acres per person (log acres/member)	0.239***	1.333***	-0.0193	-0.0526**
	(0.012)	(0.025)	(0.0198)	(0.0232)
Purestand cropping (binary)	0.031*	-0.018	0.0185	0.0426***
	(0.018)	(0.014)	(0.018)	(0.0147)
Irrigated plot (binary)	-0.151***	0.147*	-0.0457	0.0466
	(0.041)	(0.078)	(0.0699)	(0.0669)
Soil quality perceived as good (binary)	0.003	0.0119	0.0387**	0.0274
	(0.015)	(0.019)	(0.0189)	(0.0171)
Steep slope (binary)	0.016	0.046*	0.0494	0.0592**
	(0.018)	(0.026)	(0.0532)	(0.0292)
Plot is owned (binary)	-0.022	0.017	0.0550*	0.0922***
	(0.019)	(0.021)	(0.0299)	(0.0331)
Number of crops grown (#)	-0.076***	0.026*	0.0438**	0.0534***
	(0.013)	(0.014)	(0.0194)	(0.0106)
Year	Yes	Yes	Yes	Yes
Household	Yes	Yes	Yes	Yes
Crops	Yes	Yes	Yes	Yes
Observations	28537	32649	9197	16582
R^2 (within)	0.273	0.384	0.339	0.304
Joint significance F-stat	37.71	1873.94	2.370	4.080
Joint significance p-value	0.000	0.000	0.0100	0 .0000

Notes: This table illustrates the relationship between measurement error and household-level and plot-level observables. The F-statistic tests for the joint test of significance for all covariates except GPS-measured plot size. All regressions include household, year and crop fixed effects, and plot and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2 Parcel-level Attrition Due to Missing GPS Measurement

	Retained	Dropped	Difference
Ethiopia			
Head is male (binary)	0.849	0.876	-0.028***
Age of household head (years)	47.403	45.267	2.12***
Head is literate (binary)	0.428	0.482	-0.054**
Household size (# members)	5.639	5.740	-0.101***
Acres per person (log acres/member)	-0.919	-0.795	-0.123***
Self-reported parcel size (log acres)	-1.676	-1.442	-0.234***
Pure stand cropping (binary)	0.765	0.847	-0.082***
Irrigated plot (binary)	0.035	0.029	0.006
Soil quality perceived as good (binary)	0.292	0.178	0.114***
Steep slope (binary)	0.114	0.076	0.038***
Plot is owned (binary)	0.864	0.734	0.129***
Number of crops grown (# crops)	1.247	1.079	0.168***
Plot grows barley	0.052	0.056	-0.004
Plot grows maize	0.065	0.063	0.002
Plot grows sorghum	0.168	0.140	0.028***
Plot grows teff	0.143	0.107	0.037***
Plot grows wheat	0.122	0.130	-0.009
Plot grows horse beans	0.076	0.095	-0.019***
Plot grows chat	0.035	0.044	-0.009**
Plot grows coffee	0.050	0.015	0.035***
Plot grows enset	0.083	0.047	0.035***
Malawi			
Head is male (binary)	0.757	0.780	-0.023***
Age of household head (years)	44.090	39.916	4.175***
Head is literate (binary)	0.137	0.303	-0.166***
Household size (# members)	4.905	4.257	0.649***
Acres per person (log acres/member)	-0.670	-0.750	0.141***
Self-reported parcel size (log acres)	-0.262	-0.119	-0.141***
Pure stand cropping (binary)	0.713	0.854	-0.141***
Irrigated plot (binary)	0.006	0.200	-0.194***
Soil quality perceived as good (binary)	0.487	0.476	0.010
Steep slope (binary)	0.103	0.127	-0.024***
Plot is owned (binary)	0.843	0.682	0.161***
Number of crops grown (# crops)	1.569	1.084	0.485***
Plot grows local maize	0.341	0.056	0.286***
Plot grows hybrid maize	0.311	0.080	0.231***
Plot grows recycled hybrid maize	0.062	0.0018	0.044***
Plot grows groundnuts	0.074	0.011	0.062***
Plot grows sorghum	0.063	0.006	0.056***
Plot grows beans	0.078	0.023	0.055***
Plot grows soya beans	0.051	0.013	0.038***
Plot grows pigeon pea	0.198	0.021	0.177***
Plot grows nkhwani	0.146	0.034	0.111***

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Parcel-level Attrition Due to Missing GPS Measurement

	Retained	Dropped	Difference
Tanzania			
Head is male (binary)	0.77	0.79	-0.01***
Age of household head (years)	49.18	47.95	1.22***
Head is literate (binary)	0.70	0.73	-0.03***
Household size (# members)	5.95	5.77	0.19***
Acres per person (log acres/member)	-0.22	-0.34	0.12***
Self-reported parcel size (log acres)	0.27	0.25	0.02
Purestand cropping	0.47	0.54	-0.07***
Irrigated plot (binary)	0.02	0.02	0.00
Soil quality perceived as good (binary)	0.40	0.39	0.01**
Steep slope (binary)	0.03	0.03	-0.00
Plot is owned (binary)	0.86	0.75	0.12***
Number of crops grown (# crops)	1.69	1.51	0.18***
Plot grows beans	0.19	0.15	0.04***
Plot grows maize	0.67	0.52	0.14***
Plot grows sunflower	0.05	0.04	0.01***
Plot grows sorghum	0.08	0.07	0.01
Plot grows cowpeas	0.05	0.04	0.01***
Plot grows millet	0.02	0.02	0.00
Plot grows gnuts	0.11	0.08	0.02***
Plot grows rice	0.12	0.22	-0.10***
Plot grows simsim	0.02	0.03	-0.01***
Plot grows sweet potato	0.08	0.06	0.03***
Plot grows pigeon peas	0.06	0.04	0.01***
Plot grows cotton	0.04	0.02	0.01***
Plot grows bambara nuts	0.02	0.02	-0.00
Uganda			
Head is male (binary)	0.72	0.72	-0.00
Age of household head (years)	47.20	45.73	1.47***
Head is literate (binary)	0.65	0.68	-0.03***
Household size (# members)	7.02	7.21	-0.20***
Acres per person (log acres/member)	-1.04	-1.07	0.03***
Self-reported parcel size (log acres)	-0.03	-0.04	0.01
Purestand cropping (binary)	0.44	0.58	-0.13***
Irrigated plot (binary)	0.02	0.01	0.01***
Soil quality perceived as good (binary)	0.60	0.65	-0.05***
Steep slope (binary)	0.11	0.13	-0.03***
Plot is owned (binary)	0.89	0.64	0.24***
Number of crops grown (# crops)	2.48	1.93	0.54***
Plot grows maize	0.33	0.30	0.03***
Plot grows millet	0.06	0.06	0.00
Plot grows sorghum	0.08	0.07	0.01***
Plot grows beans	0.36	0.33	0.03***
Plot grows gnuts	0.10	0.10	-0.00
Plot grows sweet potato	0.22	0.18	0.04***
Plot grows cassava	0.36	0.32	0.05***
Plot grows matooke	0.36	0.19	0.17***
Plot grows beer banana	0.06	0.02	0.04***
Plot grows sweet banana	0.03	0.02	0.02***
Plot grows coffee	0.14	0.06	0.09***
Plot left fallow	0.04	0.02	0.01***

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3 Input use rates (for those plots that received the input)

	Input applied		Mean input intensity		Median input intensity	
	# plots	% plots	(Self-report)	(GPS)	(Self-report)	(GPS)
Ethiopia						
Labor (days/acre)	27,009	92	88.91	106.65	26.22	26.61
Fertilizer (kg/acre)	8,279	28	172.57	123.24	56.11	50.90
Improved seed (kg/acre)	11,259	39	87.05	74.05	23.51	21.59
Malawi						
Labor (days/acre)	29,725	91	250.48	292.08	168.00	180.00
Fertilizer (kg/acre)	18,732	57	83.93	99.02	50.00	50.00
Improved Seed (kg/acre)	4,447	14	26.34	32.42	12.00	12.75
Pest/herbicide (kg/acre)	876	3	26.44	29.45	1.00	0.98
Tanzania						
Labor (days/acre)	11,649	84	49.14	69.12	30.00	32.67
Fertilizer (kg/acre)	1,455	11	214.60	263.64	40.00	42.86
Improved Seed (value/acre)	1,552	11	15.85	19.66	8.00	9.13
Pest/herbicide (value/acre)	1,228	9	14.38	16.28	6.25	6.67
Uganda						
Labor (days/acre)	16,506	84	193.56	227.02	102.00	106.67
Fertilizer (kg/acre)	243	1	27.57	28.54	7.78	7.69
Improved Seed (value/acre)	1,345	7	16.99	20.05	8.33	8.57
Pest/herbicide (value/acre)	914	5	18.55	30.17	7.50	7.45

Notes: For Tanzania and Uganda improved seed and pesticide/herbicide application is given in terms of value purchased per acre. All other inputs are in kg/acre, except for labor which is listed in days/acre.

Table A4 Measurement error and plot size, unconditional regressions

	Ethiopia		Malawi		Tanzania		Uganda	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ln (GPS plot size):	-0.301***		-0.458***		-0.334***		-0.299***	
	(0.010)		(0.010)		(0.00705)		(0.0121)	
ln (SR plot size):		0.097***		0.183***		-0.0293***		0.0603***
		(0.008)		(0.013)		(0.00849)		(0.0103)
Observations	29182	29182	32685	32685	13847	13847	19738	19738
R^2	0.241	0.020	0.362	0.038	0.354	0.00176	0.260	0.00788

Notes: Standard errors, clustered at the household level, are given in parentheses. No controls or fixed effects are included. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5 Measurement error and input use intensity, including non-linear terms

	(1) Labor	(2) Seeds	(3) Fertilizer	(4) Pesticides
Ethiopia				
ln (GPS plot size)	-0.831***	-0.240***	-0.321***	
	(0.008)	(0.018)	(0.039)	
ln (GPS plot size)^2	-0.010***	0.040***	0.021*	
	(0.001)	(0.005)	(0.011)	
Measurement error	**0.073*** **	**0.169*** **	**0.228*** **	
	(0.008)	(0.020)	(0.026)	
Observations	26474	10959	8092	
R^2	0.783	0.461	0.254	
Malawi				
ln (GPS plot size)	-0.497***	-0.439***	-0.544***	-0.293**
	(0.013)	(0.097)	(0.020)	(0.132)
ln (GPS plot size)^2	-0.010*	-0.043	-0.024***	0.107
	(0.005)	(0.029)	(0.009)	(0.075)
Measurement error	**0.292*** **	**0.334** **	**0.246*** **	**0.229** **
	(0.014)	(0.124)	(0.021)	(0.116)
Observations	29683	4447	18723	871
R^2	0.549	0.289	0.475	0.522
Tanzania				
ln (GPS plot size)	-0.488***	-0.577***	-0.363***	-0.664***
	(0.0124)	(0.0447)	(0.0498)	(0.0638)
ln (GPS plot size)^2	-0.0297***	-0.00857	-0.0689***	0.00487
	(0.00447)	(0.0142)	(0.0160)	(0.0165)
Measurement error	**0.266*** **	**0.172*** **	**0.344*** **	**0.300*** **
	(0.0194)	(0.0660)	(0.0762)	(0.0875)
Observations	9175	1487	1305	949
R^2	0.511	0.484	0.368	0.521
Uganda				
ln (GPS plot size)	-0.699***	-0.730***	-0.450**	-0.604***
	(0.0160)	(0.0909)	(0.203)	(0.0835)
ln (GPS plot size)^2	-0.0193***	0.0269	0.206	-0.0186
	(0.00479)	(0.0308)	(0.135)	(0.0334)
Measurement error	**0.113*** **	**0.131**	**-0.187**	**0.305*** **
	(0.0173)	(0.0955)	(0.258)	(0.0973)
Observations	16164	1335	241	906
R^2	0.395	0.320	0.614	0.322

Notes: This table provides relationships between alternative indicators of measurement error and input use intensity. We construct relative overestimations and underestimations in self-reported plot sizes, for those over and under-estimated plot sizes. All regressions include household, year and crop fixed effects, and plot and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6 Input intensity and measurement error decomposed into over-/under-estimate

	(1) Labor	(2) Seeds	(3) Fertilizer	(4) Pesticides
Ethiopia				
ln (GPS-measured plot size)	-0.790***	-0.368***	-0.378***	
	(0.005)	(0.013)	(0.018)	
Overestimation (% area)	**0.022***	**0.305*****	**0.289*****	
	(0.011)	(0.030)	(0.047)	
Underestimation (% area)	**-0.122*****	**-0.118*****	**-0.192*****	
	(0.012)	(0.032)	(0.033)	
Observations	26458	10975	8089	
R^2	0.783	0.445	0.260	
Malawi				
ln (GPS-measured plot size)	-0.480***	-0.376***	-0.517***	-0.351***
	(0.011)	(0.098)	(0.019)	(0.157)
Overestimation (% area)	**0.299*****	**0.300***	**0.244*****	**0.437***
	(0.019)	(0.157)	(0.032)	(0.237)
Underestimation (% area)	**-0.273*****	**-0.348****	**-0.226*****	**-0.060**
	(0.024)	(0.172)	(0.041)	(0.252)
Observations	28380	4307	17934	835
R^2	0.550	0.289	0.478	0.504
Tanzania				
ln (GPS-measured plot size)	-0.510***	-0.599***	-0.373***	-0.677***
	(0.0132)	(0.0468)	(0.0520)	(0.0683)
Overestimation (% area)	**0.200*****	**0.0105**	**0.161**	**0.310****
	(0.0289)	(0.0922)	(0.103)	(0.146)
Underestimation (% area)	**-0.270*****	**-0.290*****	**-0.401*****	**-0.244****
	(0.0301)	(0.103)	(0.106)	(0.110)
Observations	8937	1458	1265	930
R^2	0.509	0.491	0.345	0.536
Uganda				
ln (GPS-measured plot size)	-0.692***	-0.699***	-0.357*	-0.580***
	(0.0167)	(0.0924)	(0.197)	(0.0848)
Overestimation (% area)	**0.0727*****	**0.345****	**0.705**	**0.266**
	(0.0270)	(0.158)	(0.683)	(0.173)
Underestimation (% area)	**-0.151*****	**0.00883**	**0.297**	**-0.427*****
	(0.0264)	(0.165)	(0.387)	(0.147)
Observations	15760	1302	227	874
R^2	0.397	0.323	0.616	0.335

Notes: This table provides relationships between alternative indicators of measurement error and input use intensity. We construct relative overestimations and underestimations in self-reported plot sizes, for those over and under-estimated plot sizes. All regressions include household, year and crop fixed effects, and plot and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7 Measurement error and the extensive margin of input use

	(1) Improved seed (dummy)	(2) Fertilizer (dummy)	(3) Pesticide (dummy)
Ethiopia			
ln (GPS-measured plot size)	0.051***	0.054***	0.024***
	(0.002)	(0.002)	(0.002)
Measurement error	**0.008****	**0.020*****	**0.007*****
	(0.004)	(0.003)	(0.002)
Observations	28481	28537	28537
R^2	0.592	0.219	0.173
Malawi			
ln (GPS-measured plot size)	-0.059**	0.157***	0.024***
	(0.028)	(0.007)	(0.003)
Measurement error	**-0.061***	**0.080*****	**0.011*****
	(0.034)	(0.008)	(0.003)
Observations	7983	32649	32649
R^2	0.269	0.378	0.048
Tanzania			
ln (GPS-measured plot size)	0.0162***	0.0236***	0.0252***
	(0.00565)	(0.00491)	(0.00460)
Measurement error	**0.0146***	**0.00980**	**0.0129****
	(0.00756)	(0.00693)	(0.00561)
Observations	7987	9197	9197
R^2	0.0974	0.0354	0.0943
Uganda			
ln (GPS-measured plot size)	0.0117***	0.00485**	0.0124***
	(0.00416)	(0.00204)	(0.00354)
Measurement error	**0.00889****	**0.00226**	**0.00730****
	(0.00429)	(0.00195)	(0.00367)
Observations	16582	16571	16571
R^2	0.0479	0.0131	0.0349

Notes: This table provides relationships between measurement error and the extensive margin of input use. We rely on binary (indicator) variables showing whether a farmer has applied a specific input in a specific plot. All regressions include household, year and crop fixed effects, and plot and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8 Measurement error by direction and the extensive margin of input use

	(1) Improved seed (dummy)	(2) Fertilizer (dummy)	(3) Pesticide (dummy)
Ethiopia			
ln (GPS-measured plot size)	0.049***	0.056***	0.024***
	(0.002)	(0.002)	(0.002)
Overestimation (% area)	**-0.002**	**0.030*** **	**0.009*** **
	(0.005)	(0.004)	(0.003)
Underestimation (% area)	**-0.025*** **	**-0.006**	**-0.004**
	(0.006)	(0.006)	(0.005)
Observations	28464	28520	28520
R^2	0.592	0.219	0.173
Malawi			
ln (GPS-measured plot size)	-0.060**	0.158***	0.026***
	(0.030)	(0.007)	(0.003)
Overestimation (% area)	**-0.047**	**0.082*** **	**0.011** **
	(0.048)	(0.011)	(0.005)
Underestimation (% area)	**0.082**	**-0.073*** **	**-0.011* **
	(0.050)	(0.013)	(0.006)
Observations	7682	31220	31220
R^2	0.269	0.380	0.050
Tanzania			
ln (GPS-measured plot size)	0.0135**	0.0219***	0.0256***
	(0.00584)	(0.00491)	(0.00487)
Overestimation (% area)	**0.0133**	**0.0107**	**0.0120**
	(0.0108)	(0.0101)	(0.00805)
Underestimation (% area)	**-0.0102**	**-0.00323**	**-0.0155* **
	(0.0121)	(0.0110)	(0.00867)
Observations	7778	8959	8959
R^2	0.0973	0.0348	0.0938
Uganda			
ln (GPS-measured plot size)	0.0113***	0.00466**	0.0106***
	(0.00428)	(0.00200)	(0.00351)
Overestimation (% area)	**0.00721**	**-0.00104**	**-0.00442**
	(0.00641)	(0.00242)	(0.00551)
Underestimation (% area)	**-0.00901**	**-0.00513**	**-0.0166*** **
	(0.00658)	(0.00367)	(0.00524)
Observations	16175	16164	16164
R^2	0.0480	0.0134	0.0352

Notes: This table provides relationships between alternative indicators of measurement error (overestimations and underestimations) and extensive margin of input use decisions. All regressions include household, year and crop fixed effects, and plot and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9 Equation 19 estimated combining intensive and extensive margins

	(1) Labor	(2) Seeds	(3) Fertilizer	(4) Pesticides
Ethiopia				
log(GPS plot size)	-0.423***	0.085***	0.194***	
	(0.013)	(0.011)	(0.011)	
Measurement error	0.125***	0.049***	0.120***	
	(0.020)	(0.018)	(0.017)	
Observations	28537	28537	28537	
R^2	0.199	0.534	0.198	
Malawi				
log(GPS plot size)	-0.453***	-0.496***	0.505***	0.013*
	(0.021)	(0.027)	(0.034)	(0.007)
Measurement error	0.266***	0.141***	0.477***	0.009
	(0.025)	(0.032)	(0.043)	(0.007)
Observations	32649	32649	32649	32649
R^2	0.151	0.314	0.207	0.014
Tanzania				
log(GPS plot size)	-0.492***	0.0122	0.0425**	0.183***
	(0.0125)	(0.0536)	(0.0215)	(0.0423)
Measurement error	0.232***	0.108	0.0916***	0.134**
	(0.0202)	(0.0707)	(0.0333)	(0.0524)
Observations	9197	7987	9196	9197
R^2	0.481	0.0819	0.0262	0.0872
Uganda				
log(GPS plot size)	-0.653***	0.0244	0.00881	0.0994***
	(0.0221)	(0.0379)	(0.00698)	(0.0346)
Measurement error	0.110***	0.0614	0.0134*	0.0726**
	(0.0239)	(0.0388)	(0.00740)	(0.0369)
Observations	16571	16582	16571	16571
R^2	0.248	0.0458	0.0112	0.0317

Notes: This table estimates Equation 19, but jointly estimating the extensive and intensive margin input decisions; input application is estimated using an inverse hyperbolic sine transformation. All regressions include household, year and crop fixed effects and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10 Equation 19 estimated at the farm-aggregate level

	(1) Labor	(2) Seeds	(3) Fertilizer	(4) Pesticides
Ethiopia				
log(GPS farm-total area)	-0.489***	-0.570***	-0.351***	
	(0.025)	(0.075)	(0.047)	
Farm-total measurement error	0.232***	0.035	0.356***	
	(0.039)	(0.121)	(0.059)	
Observations	6108	2881	2964	
R^2	0.250	0.756	0.222	
Malawi				
log(GPS farm-total area)	-0.372***	-0.402***	-0.476***	-0.747***
	(0.010)	(0.033)	(0.013)	(0.150)
Farm-total measurement error	0.329***	0.220***	0.243***	-0.043
	(0.015)	(0.044)	(0.018)	(0.202)
Observations	18797	3579	14105	779
R^2	0.368	0.244	0.332	0.127
Tanzania				
log(GPS farm-total area)	-0.589***	-0.748***	-0.573***	-0.384
	(0.0403)	(0.201)	(0.201)	(0.280)
Farm-total measurement error	0.0631***	0.00438	0.0365	0.155*
	(0.0167)	(0.115)	(0.0364)	(0.0922)
Observations	4862	1055	854	751
R^2	0.367	0.385	0.283	0.309
Uganda				
log(GPS farm-total area)	-0.640***	-0.440***	-1.571***	-1.023***
	(0.0280)	(0.146)	(0.460)	(0.163)
Farm-total measurement error	0.0541***	0.0693	-0.466	0.126***
	(0.0118)	(0.0529)	(0.288)	(0.0461)
Observations	11135	1167	215	813
R^2	0.346	0.251	0.698	0.216

Notes: This table estimates Equation 19 at the farm, rather than plot, level. GPS farm-total area is the total area of all GPS-measured plots within farm and round. Farm-level input intensity is given by log(farm-aggregate input quantity / GPS farm-total area). Farm-total measurement error is given by log(GPS farm-total area)-log(self-reported farm-total area). All regressions include household, year and crop fixed effects and household controls. For all rows, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1 Linearity of the Relationship Between Measurement Error and Log GPS Plot Size

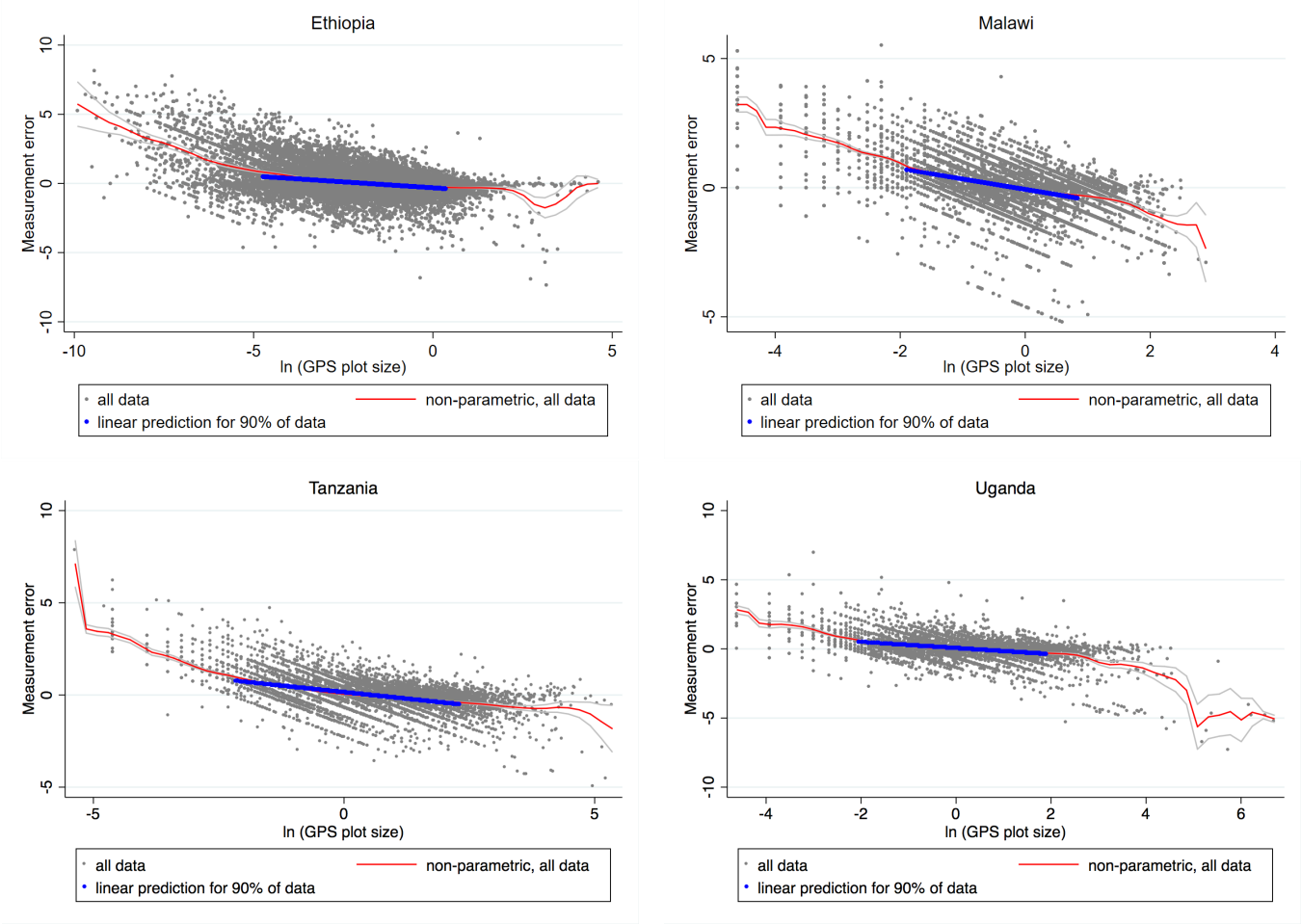
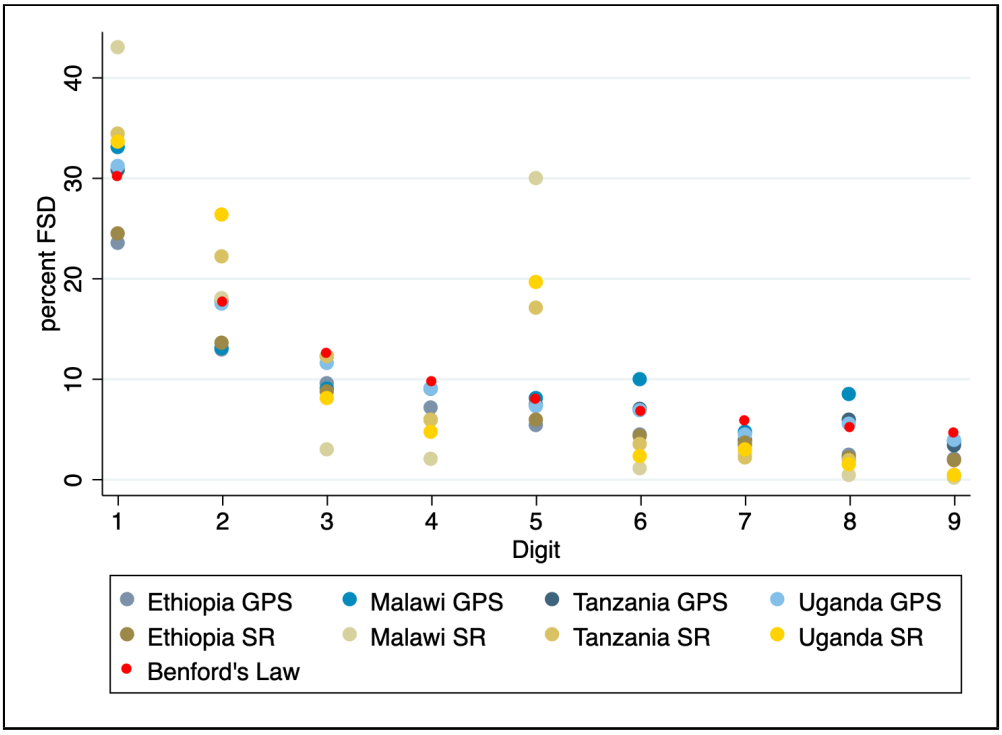
Figure A2 GPS vs. Self-reported Plot Size Adherence to Benford's Law